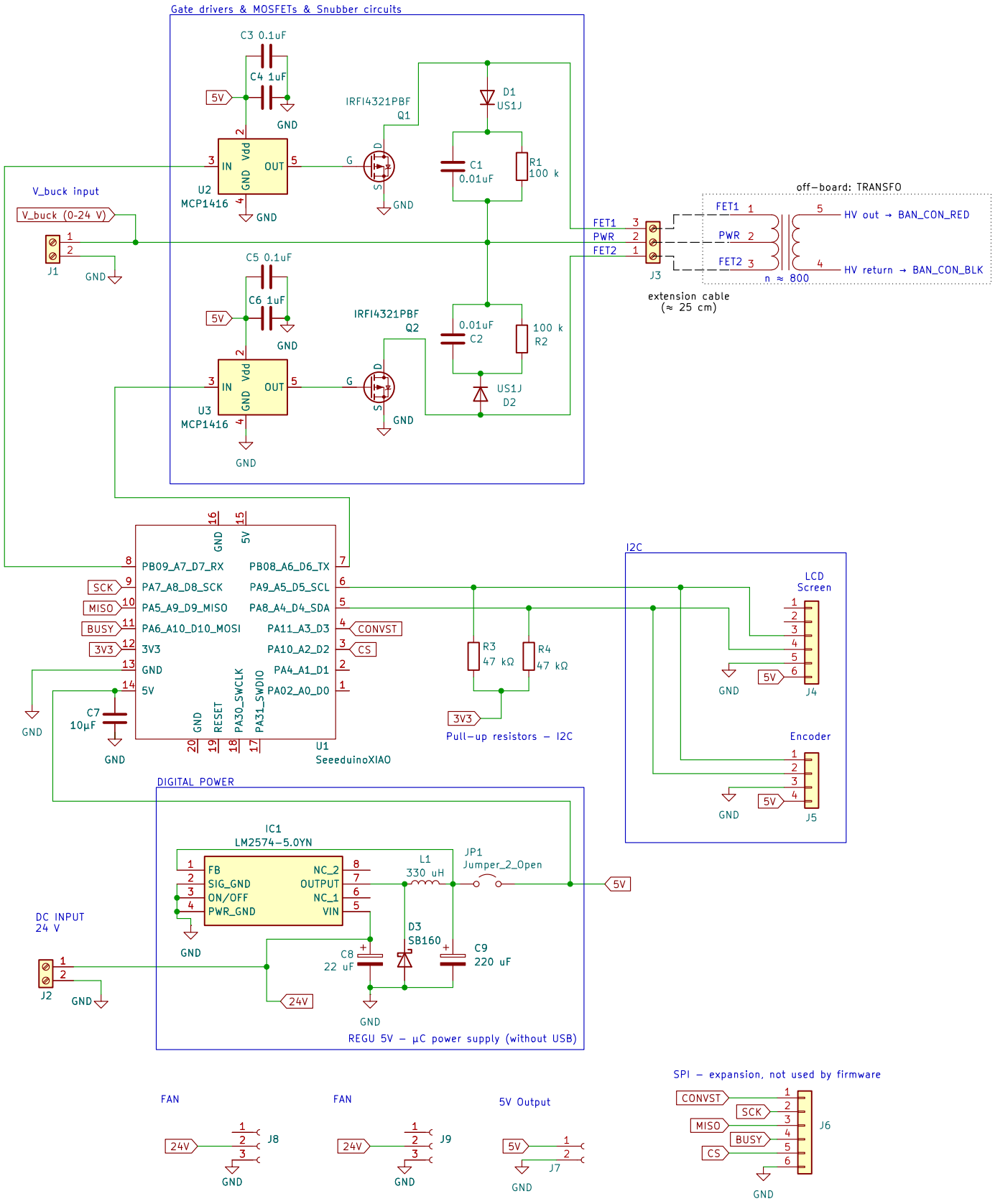

Gate drivers & MOSFETs & Snubber circuits
C3 0.1uF
C4 1uF
5V
GND
IRFI4321PBF
Q1
D1
US1J
C1
0.01uF
R1
100 k
U2
MCP1416
IN
GND
Vdd
OUT
G
D
S
V_buck input
V_buck (0-24 V)
J1
C5 0.1uF
C6 1uF
IRFI4321PBF
Q2
0.01uF
C2
100 k
R2
US1J
D2
U3
MCP1416
off-board: TRANSFO
FET1
PWR
FET2
J3
HV out → BAN_CON_RED
HV return → BAN_CON_BLK
n ≈ 800
extension cable
(≈ 25 cm)
PB09_A7_D7_RX
PB08_A6_D6_TX
PA7_A8_D8_SCK
PA9_A5_D5_SCL
PA5_A9_D9_MISO
PA8_A4_D4_SDA
PA6_A10_D10_MOSI
PA11_A3_D3
PA10_A2_D2
PA4_A1_D1
PA02_A0_D0
3V3
GND
5V
SCK
MISO
BUSY
CONVST
CS
RESET
PA30_SWCLK
PA31_SWDIO
C7
10µF
U1
SeeeduinoXIAO
R3
47 kΩ
R4
47 kΩ
Pull-up resistors – I2C
I2C
LCD
Screen
J4
Encoder
J5
DIGITAL POWER
IC1
LM2574-5.0YN
FB
SIG_GND
ON/OFF
PWR_GND
NC_2
OUTPUT
NC_1
VIN
L1
330 uH
JP1
Jumper_2_Open
D3
SB160
C8
22 uF
C9
220 uF
24V
DC INPUT
24 V
J2
REGU 5V – µC power supply (without USB)
FAN
J8
FAN
J9
5V Output
J7
SPI – expansion, not used by firmware
J6